Joel Rodriguez
Fall 2026
CIS-245
Sed Lab

1. Change the name Jo to Josephine - **COMMAND:** *sed -n "s/^Jo /Josephine /p" SedLab*

```
joelrodriguez@localhost:~/Documents$ sed -n "s/^Jo /Josephine /p" SedLab
Josephine March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
```

Source: https://www.grymoire.com/Unix/Sed.html#uh-31

This command works because sed uses the s command to substitute Jo with Josephine. The first part tells sed to look in the beginning of each line and what name to look for also, the second part tells it what to replace it with. The p prints the updated line while -n prevents the other lines from being displayed.

2. Delete the last 5 lines. - **COMMAND:** *sed '49, $d' SedLab*

```
joelrodriguez@localhost:~/Documents$ sed '49, $d' SedLab
Huckleberry Finn:238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100
Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600
Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
joelrodriguez@localhost:~/Documents$
```

Source: https://linuxize.com/post/sed-delete-lines/

This command works because sed uses the line numbers to determine which lines to delete. First I used the *wc -l* commandto find out how many lines the SedLab had. There are 53 lines. Since $ means the end, “49,$” is selecting everything from line 49 until the end of the file and the d deletes those lines from the output.

3. Print lines 3-15. - **COMMAND**: *sed -n “3,15 p” SedLab*

```
joelrodriguez@localhost:~/Documents$ sed -n "3,15 p" SedLab
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500

Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400

Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700

Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900

Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100

Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100

Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
```

Source: https://www.grymoire.com/Unix/Sed.html#uh-31

This command works because -n stops sed from automatically printing every line. The 3,15 selects the range of lines that were asked for, and p tells sed to only print those specific lines. This allows us to display lines 3-15 without showing the rest of the file.

4. Delete lines for people who live in California - **Command**: *sed “/CA /d” SedLab*

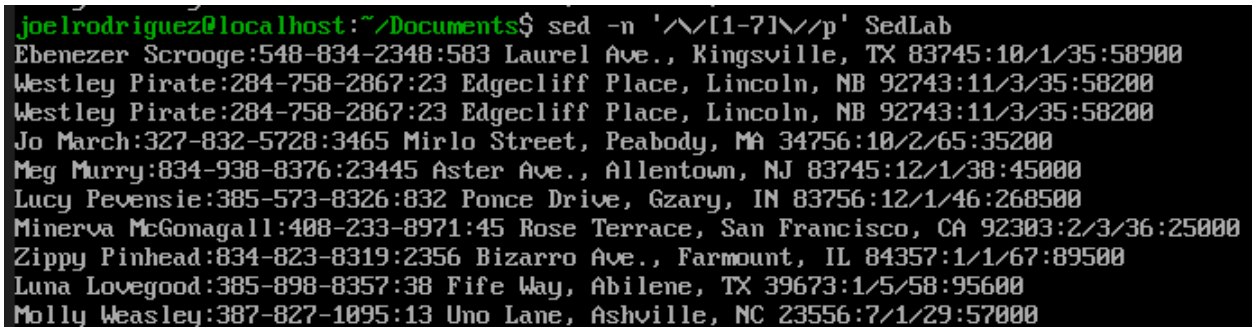


Source: https://linuxize.com/post/sed-delete-lines/

This command works because sed searches for lines containing CA (California) in the file. I made sure to leave a space after the CA so sed can search for the

state initials because in the past I have had the issues where command will search for words that had the letters in it so the space will differentiate the states from the words that ca in it. The d will delete all the lines that are from California.

5. Print all lines where the birthdays are in the first week of the month. Be careful of the dates for birthdays, the format is MM/DD/YY - **COMMAND:** sed -n '/\/[1-7]\//p' Sedlab



Source: https://www.grymoire.com/Unix/Sed.html#uh-31

This command works because sed searches for birthdays that fall in the first week of every month. Week is 7 days, so we're looking for days that fall on the day on the 1st through the 7th. The regular expression uses [1-7] to match those days. The forward slashes help identify the date format. The p only prints the lines containing the matching birthdays and the -n makes sure sed doesn't print every other line.

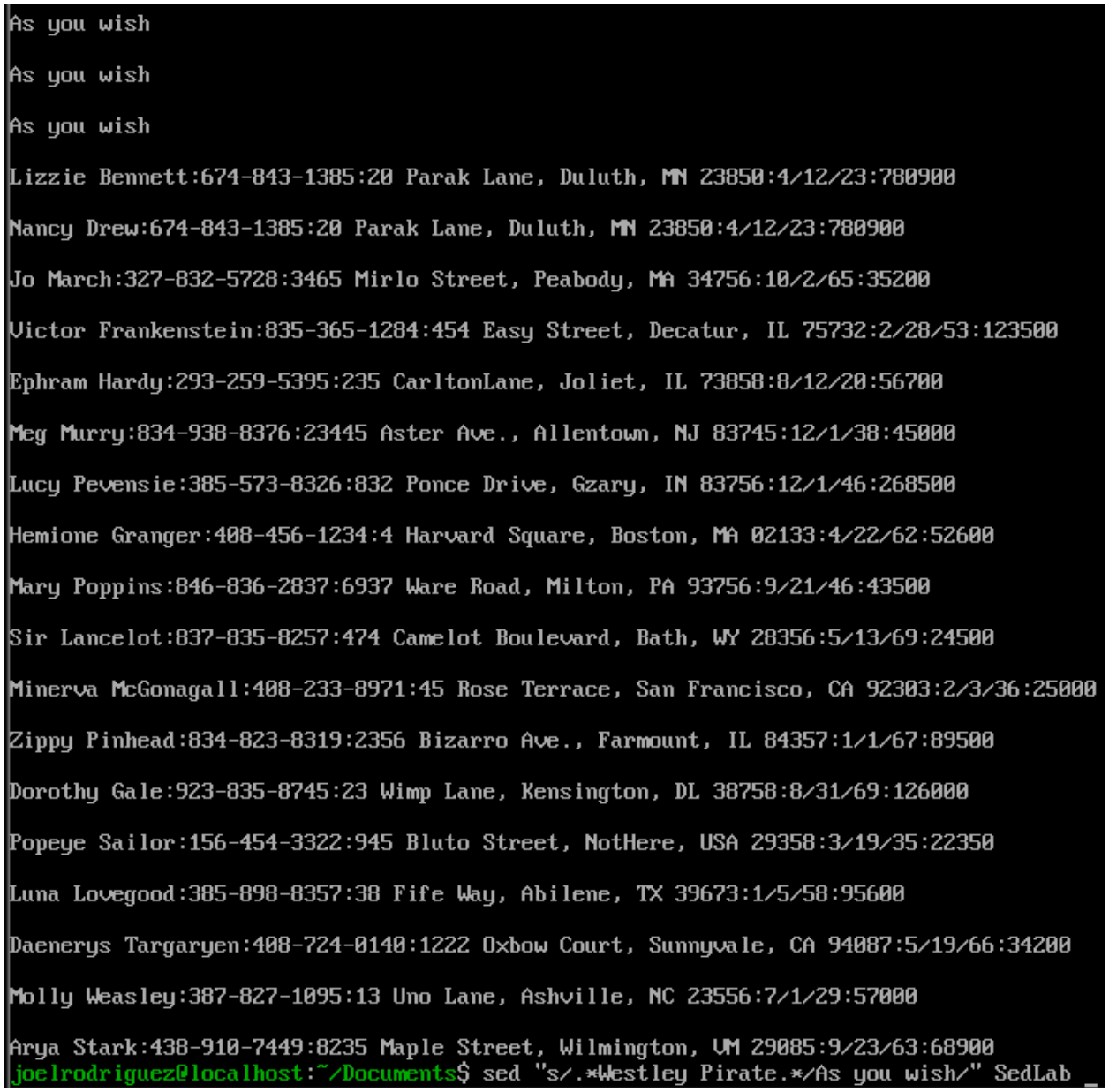
6. Append three asterisks (*) to the end of lines starting with Sir -
COMMAND*: sed "/^Sir/a ***" SedLab*



Source: https://nickjanetakis.com/blog/insert-text-above-below-a-match-with-sed

This command works because sed uses ^Sir to find lines that begin with Sir. The ^ makes sure it only matches Sir in the beginning of the line and nowhere else. The a tells sed to append text after each matching line and the *** is the text that is being added.

7. Replace the line containing "Westley Pirate" with the phrase "As you wish." Make sure you replace the whole line not just the name
   COMMAND: *sed “s/.*Westley Pirate.*/As you wish/” SedLab*



Source:
https://www.oreilly.com/library/view/practical-unix/078972250X/078972250X_ch19lev1sec7.html

This command works because the sed command is finding every line containing Westley Pirate and replace the entire line with “As you wish.” The s makes the substitution while “.*” matches all the characters before and after the name. This makes sure the whole line is replaced instead of just the person’s name.

8. Change Minerva McGonagall's birthday to 10/04/1935. Assume you don't know Minerva's original birthday. Use a regular expression to search for it.

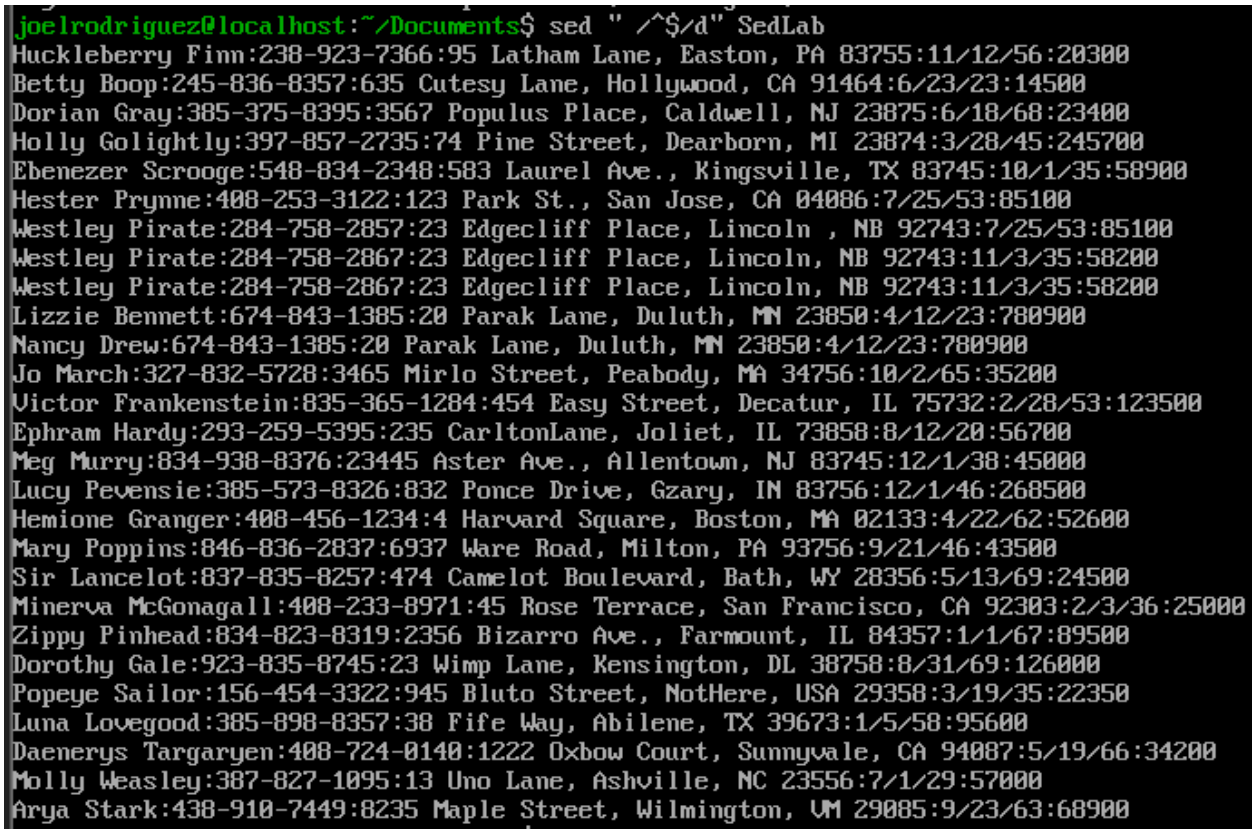
   COMMAND: sed “/Minerva McGonagall/s|[0-9]*/[0-9]*/[0-9]*:|10/04/1935:|” SedLab



   Source: https://www.grymoire.com/Unix/Sed.html#uh-5

   This command works because sed searches for the line containing Minerva McGonagall and uses s to replace her original birthday 2/3/36 with 10/04/1935. The regular expression on [0-9]* helps matches the digits without needing to know her original birthday. The forward slashes formats the date, while the colon will sed where the birthday ends.

9. Delete all blank lines. - **COMMAND**: *sed "/^$/d" SedLab*



Source: https://www.cyberciti.biz/faq/using-sed-to-delete-empty-lines/

This command works because sed uses ^$ to find empty lines. The ^ represents the beginning and $ represents the end. When these symbols are together, they match with lines that have nothing between them. The d tells sed to delete those empty lines.

10. Write a sed script that will (actual sed script, NOT just the commands on the command line) enumerate - **SCRIPT**: = **COMMAND**: *sed -f numberlines.sed SedLab*



SOURCE: https://www.grymoire.com/Unix/Sed.html#uh-48

I created a sed script called numberlines.sed using the nano text editor. Inside the script I just added “=”, this simple command tells sed to print the line

number on top of each line. So that command to run the command towards the SedLab file. This command works because the -f option tells sed to read the command from script and I won't have to type it in the command line.

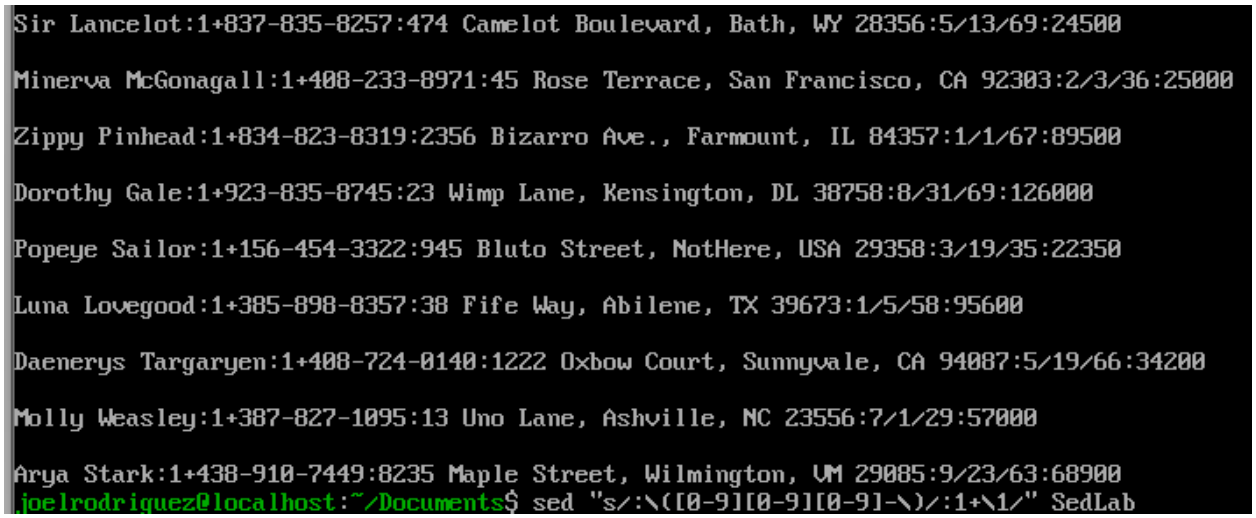
11. Insert above the first line the title - Great Literary Characters -. -
    COMMAND: *sed "/^Huckleberry/i - Great Literary Characters –" SedLab | less*



Source: https://nickjanetakis.com/blog/insert-text-above-below-a-match-with-sed

This command works because sed finds the line beginning with Huckleberry and insert the title “-Great Literary Characters-“ above it. The forward slashes surround the word we’re searching for, and the ^ makes sure Huckleberry appears at the beginning of the line. The i tells sed to insert the new text right before the matching line. Adding the less command allows us to scroll up and show the very first line so we can see the new title of the text file.

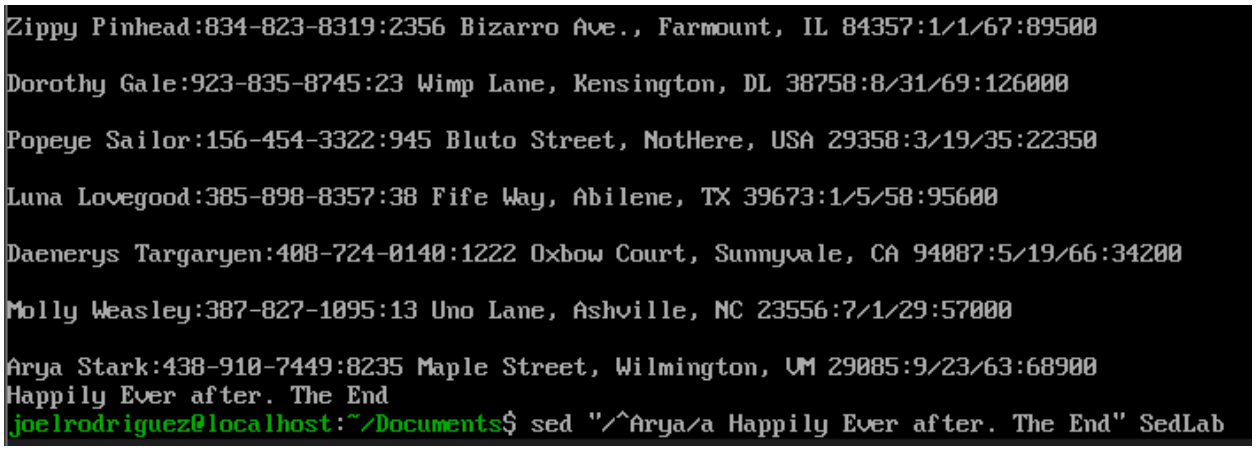
12. Print the contents of the file, but instead of the phone number starting with an area code, have it start with a 1+, then include the area code and number. For example, (603)123-1234 would turn into 1+(603)123-1234
COMMAND: *sed "s/:\([0-9] [0-9] [0-9]-\)/:1+\1/" SedLab*



Source: https://www.redhat.com/en/blog/manipulating-text-sed

This command works because sed uses s to substitute part of each phone number. The regular expression [0-9] matches digits from 0 through 9, allowing sed to find the three-digit area code. The parentheses create a group that remembers the area code and the dash after it. The \1 puts that information back into the replacement after adding 1+. This allows sed to add the prefix without removing the original number.

13. Append at the end of the file "Happily Ever after. The End" -
    COMMAND: sed “/^Arya/a Happily Ever after. The End” SedLab



Source: https://nickjanetakis.com/blog/insert-text-above-below-a-match-with-sed

This command works because sed searches for the line containing arya and uses a command to append the new text immediately after it. Since arya is the last person listed in the file the phrase “Happily Ever after. The End” appears at the end of the output. Sed will display the additional line underneath it.